Gerrymandering Metrics

2024 Election Results and the 2024 District Map

Calculating Gerrymandering: Mean-Median Score & Efficiency Gap

```
library(tidyverse)
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0
──
✔ dplyr     1.1.4     ✔ readr     2.1.5
✔ forcats   1.0.1     ✔ stringr   1.5.2
✔ ggplot2   4.0.0     ✔ tibble    3.3.0
✔ lubridate 1.9.4     ✔ tidyr     1.3.1
✔ purrr     1.1.0
── Conflicts ────────────────────────────────────────── tidyverse_conflicts()
──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```

```
g24_sov_precinct <- read_csv("~/stat133/gerrymandering-leyna-liu/data/
g24_sov_precinct.csv")
```

```
Rows: 51123 Columns: 76
── Column specification
────────────────────────────────────────────────────────
Delimiter: ","
chr  (4): SVPREC, SVPREC_KEY, ELECTION, GEO_TYPE
dbl (72): COUNTY, FIPS, ADDIST, CDDIST, SDDIST, BEDIST, TOTREG, DEMREG,
REPR...

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this
message.
```

```
# Here I created the table that has columns for total democrat votes
(d_votes); total republican votes (r_votes); total votes for republican and
democrats; the ratio of votes for democrats by using if_else, so if
total_votes > 0, then calculate the ratio, and if the total_votes is = 0, then
```

```
print NA.

# winning_party column is if d_votes > r_votes then print "D wins" if r_votes
> d_votes then "R wins", and otherwise print "Tie"

# Then I calculate the wasted votes for each party by following the framework
from the worksheet

# you need 51 votes to win, as that is just over 50, so you do d_votes - 51,
and r_votes - 51

gerrymandering_table <- g24_sov_precinct |>
    group_by(CDDIST) |>
    summarize(
        d_votes = sum(CNGDEM01, na.rm = TRUE),
        r_votes = sum(CNGREP01, na.rm = TRUE),
        .groups = "drop"
    ) |>
        mutate(
            total_votes = d_votes + r_votes,
            democrat_ratio = if_else(total_votes > 0, d_votes / total_votes,
NA),
            republican_ratio = if_else(total_votes > 0, r_votes / total_votes,
NA),
            winning_party = if_else(d_votes > r_votes, "D wins",
if_else(r_votes > d_votes, "R wins", "Tie")),
            d_wasted = if_else(winning_party == "D wins", d_votes - (51),
d_votes),
            r_wasted = if_else(winning_party == "R wins", r_votes - (51),
r_votes)
        )
```

```
# I am saving the gerrymandering_table in my data folder under gerrymandering-
leyna-liu so I can use it in DASHBOARD (part 7) of project

write_csv(gerrymandering_table, "data/gerrymandering_table.csv")
```

```
# Now I will calculate the Mean-Median value:

mean_median <- gerrymandering_table |>
    mutate(v_i = republican_ratio) |>
        summarize(mean_v = mean(v_i), med_v = median(v_i)) |>
        mutate(mean_v - med_v)

mean_median
```

```
# A tibble: 1 × 3
  mean_v med_v `mean_v - med_v`
   <dbl> <dbl>            <dbl>
1  0.384 0.373           0.0116
```

```
# So v_i is the republican vote share for each congressional district, so if
mean - median is greater than 0, that means median republican votes is lower
than the mean republican votes, and that means that median republican votes is
less than statewide average, so that means republicans are not wasting their
votes and democrats are wasting more votes.

# essentially if democrats are winning in landslides in places they already
win more easily, then they are wasting more votes. so even though democrats
are winning more congressional districts, they are wasting more votes, and are
hence less efficient.
```

```
# so here even though the gerrymandering table already has a total_votes
column, since you are feeding it through summarize you have to write total
votes again because summarize makes it so that there are only columns
total_d_wasted, total_r_wasted, and total_votes left

total_wasted <- gerrymandering_table |>
    summarize(
        total_d_wasted = sum(d_wasted, na.rm = TRUE),
        total_r_wasted = sum(r_wasted, na.rm = TRUE),
        total_votes = r_votes + d_votes
    ) |>
        mutate(eg_gap = (total_d_wasted - total_r_wasted) / total_votes)
```

```
Warning: Returning more (or less) than 1 row per `summarise()` group was
deprecated in
dplyr 1.1.0.
i Please use `reframe()` instead.
i When switching from `summarise()` to `reframe()`, remember that `reframe()`
  always returns an ungrouped data frame and adjust accordingly.
```

```
total_wasted
```

```
# A tibble: 53 × 4
   total_d_wasted total_r_wasted total_votes eg_gap
            <dbl>          <dbl>       <dbl>  <dbl>
 1       61641273       39945667    87090915  0.249
 2       61641273       39945667      318622 68.1
 3       61641273       39945667      378791 57.3
```

```
 4       61641273       39945667      421855 51.4
 5       61641273       39945667      341965 63.4
 6       61641273       39945667      348690 62.2
 7       61641273       39945667      287011 75.6
 8       61641273       39945667      295634 73.4
 9       61641273       39945667      272688 79.6
10       61641273       39945667      251099 86.4
# ℹ 43 more rows
```

Since the eg_gap is positive through all the cells in the column, that means the total_d_wasted is larger than total_r_wasted, meaning democrats wasted more votes. In addition, the mean-median is positive when v_i represents the Republican share, meaning again Democrats wasted more votes and was less efficient, even though Democrats won more districts.

2024 Election Results and the proposed 2025 District Map

PART 6: Calculating Gerrymandering Again

```r
# From data-cleaning quarto document for part 5:

final <- read_csv("data/final.csv")
```

```
Rows: 52 Columns: 8
── Column specification
────────────────────────────────────────────────────────
Delimiter: ","
chr (3): new_dist, geometry, winner
dbl (5): d_votes_total_dist, r_votes_total_dist, total, d_prop, r_prop

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this
message.
```

```r
# Mean-Median value:
mean <- mean(final$d_prop)
median <- median(final$d_prop)

new_mean_median <- mean - median

# Republican advantage since new_mean_median is a positive value

# here median represents the typical district (or middle district), and mean
represents the average democratic vote share

# so, mean = how democratic the state is overall, and median = how democratic
the typical district is
```

```
# so, when median is lower that means that the typical district is less
democratic than the overall democratic vote in the state, meaning there is
probably a few districts that are blue concentrated, which is not efficient.
```

Republican advantage since new_mean_median is a positive value

```
# Add wasted votes for both parties in final

final2 <- final |>
    mutate(
        d_wasted = if_else(
            winner == "D",
            d_votes_total_dist - 51,
            d_votes_total_dist
        ),
        r_wasted = if_else(
            winner == "R",
            r_votes_total_dist - 51,
            r_votes_total_dist
        )
    )
```

```
# calculate total wasted for both parties then calculate eg

total_d_wasted <- sum(final2$d_wasted, na.rm = TRUE)
total_r_wasted <- sum(final2$r_wasted, na.rm = TRUE)
total_votes <- sum(final2$total, na.rm = TRUE)

new_eg_value <- (total_r_wasted - total_d_wasted) / total_votes
```